

DATA VISUALIZATION · FINAL PROJECT

Driving the Electric Future

*A Global Analysis of Electric Vehicle Adoption, Infrastructure,
and Market Evolution (2011 – 2035)*

Author: Purnachandar Vallala(69854785)

Data Source: International Energy Agency (IEA) — Global EV Data by Country, 2026 Edition

Scope: 49,375 observations · 72 countries · 2011–2025 historical + 2035 projections

From Raw Data to Strategic Foresight

Three notebooks, one continuous analytical narrative — audit, explore, and forecast the global EV transition.

01

Data Audit & Preprocessing

Workbook inspection, quality assessment, duplicate removal — 49,375 IEA observations validated.

02

Exploratory Data Analysis

12 business analyses covering sales, fleet growth, technology mix, pricing, infrastructure & correlations.

03

Future Outlook & Strategic Insights

2035 scenario forecasting (CPS vs. STEPS), infrastructure needs, and executive recommendations.

A Clean, Trustworthy Analytical Foundation

49,375

Observations audited

72

Countries covered

17 yrs

Historical span (2011–2025)

9

Analytical variables

Quality Assessment Outcomes



Missing values

Limited to region_country and unit — confirmed as expected dataset characteristics (aggregates, price-only fields), not defects. Intentionally preserved.



Duplicate records

15 exact duplicate rows identified, individually verified, and removed to protect statistical integrity.



Structural validation

GEVO_EV_2026 confirmed as primary worksheet; dataset validated as consistent and analysis-ready.

Passenger EV Sales: From Niche to Mainstream

How has the global passenger car EV market evolved between 2010 and 2025?

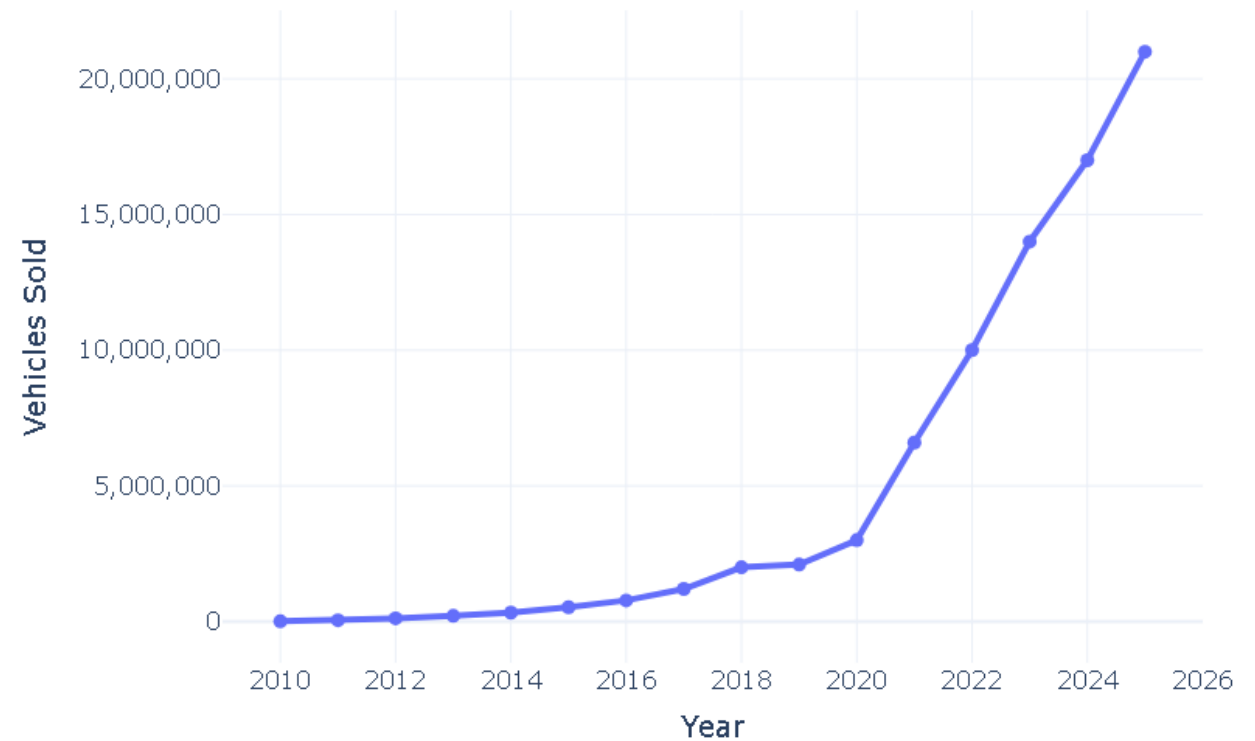
69.9% CAGR

Compound annual growth rate, 2010–2025

Key Insights

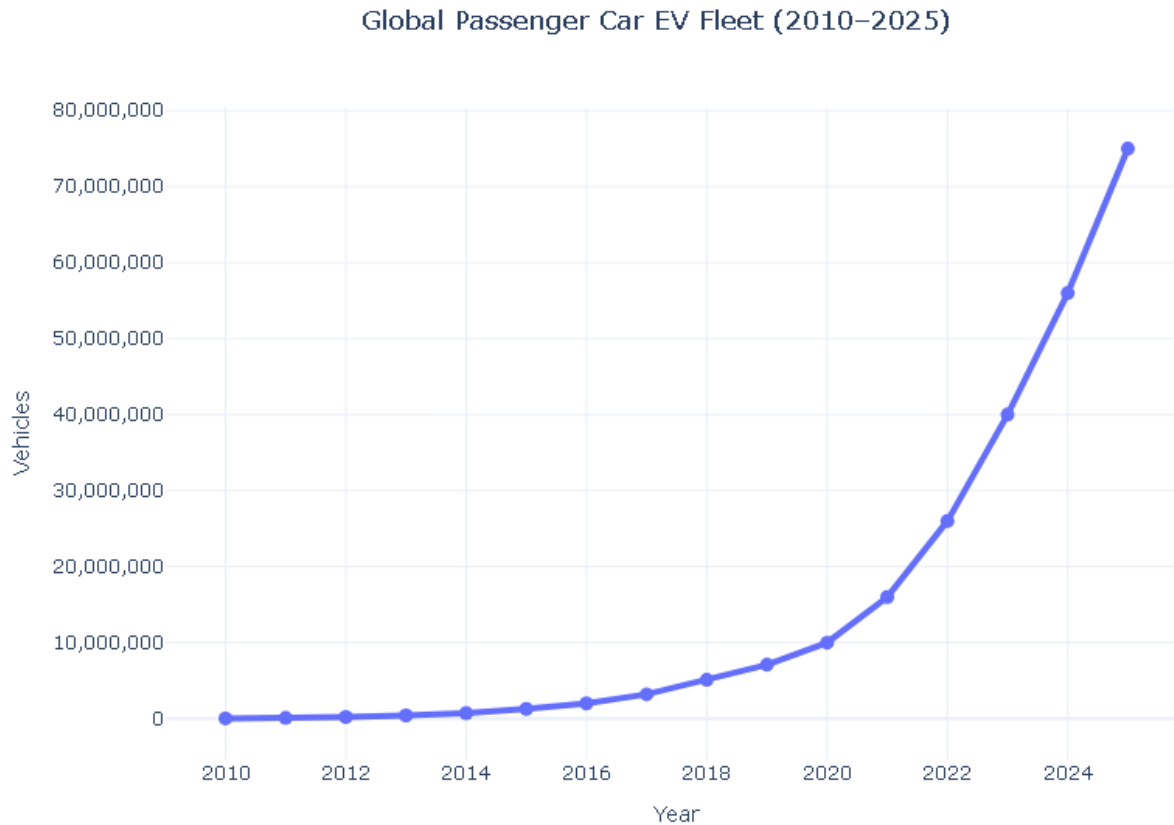
- ◆ Sales grew from 7,400 vehicles (2010) to 21 million vehicles (2025) — nearly three orders of magnitude.
- ◆ Fastest acceleration occurred after 2020, driven by policy support, falling battery costs, and expanding charging networks.
- ◆ Growth rate is moderating post-2022, but total volume keeps hitting record highs — a signal of transition from early adoption to mass commercialization.

Global Passenger Car EV Sales (2010–2025)



The Global EV Fleet Is Compounding

How has the cumulative global electric vehicle fleet evolved between 2010 and 2025?



Key Insights

- ◆ EV stock (vehicles currently on the road) shows sustained, accelerating expansion across the full study period.
- ◆ Unlike annual sales, stock reflects accumulated market penetration rather than yearly fluctuation — a truer read on adoption depth.
- ◆ The accelerating curve signals electric mobility moving from emerging technology to mainstream transportation, reinforced by infrastructure and battery innovation.

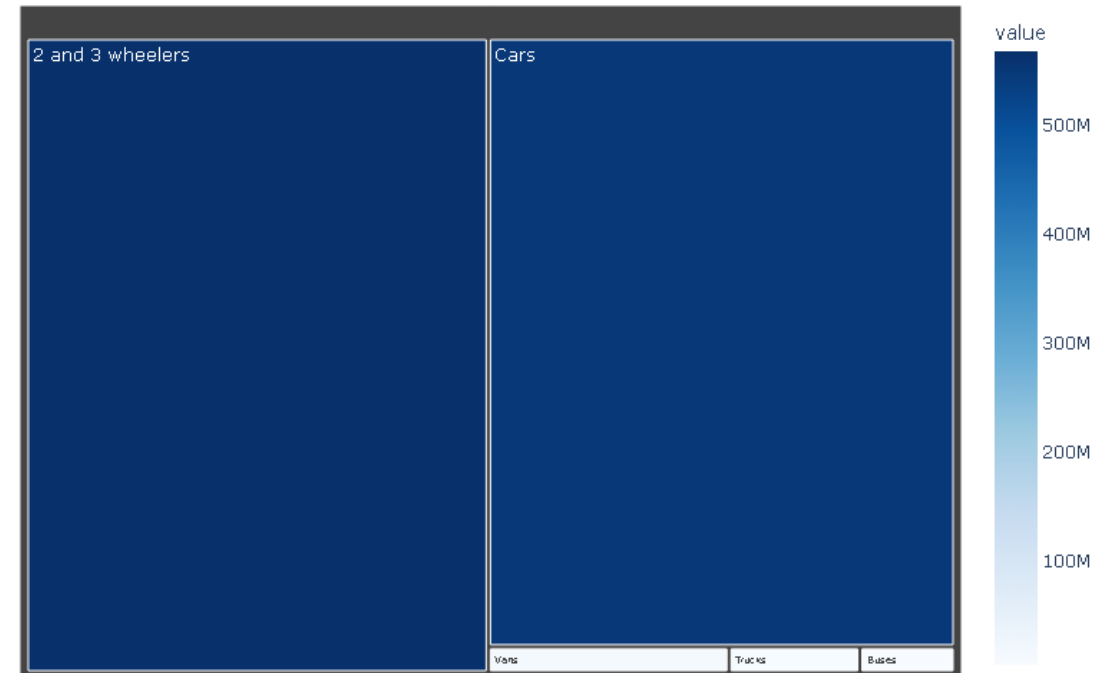
Electrification Is Spreading Beyond Passenger Cars

How has EV market composition evolved across vehicle segments and powertrains?

Key Insights

- ◆ Passenger cars remain the dominant share of global EV sales throughout the period.
- ◆ Two- and three-wheelers form the second-largest electrified segment, led by emerging markets.
- ◆ Vans and buses show steady growth from commercial logistics and public-transport electrification; trucks are earlier-stage but accelerating.
- ◆ Market diversification has broadened sharply since 2018 — electrification is no longer a passenger-car-only story.

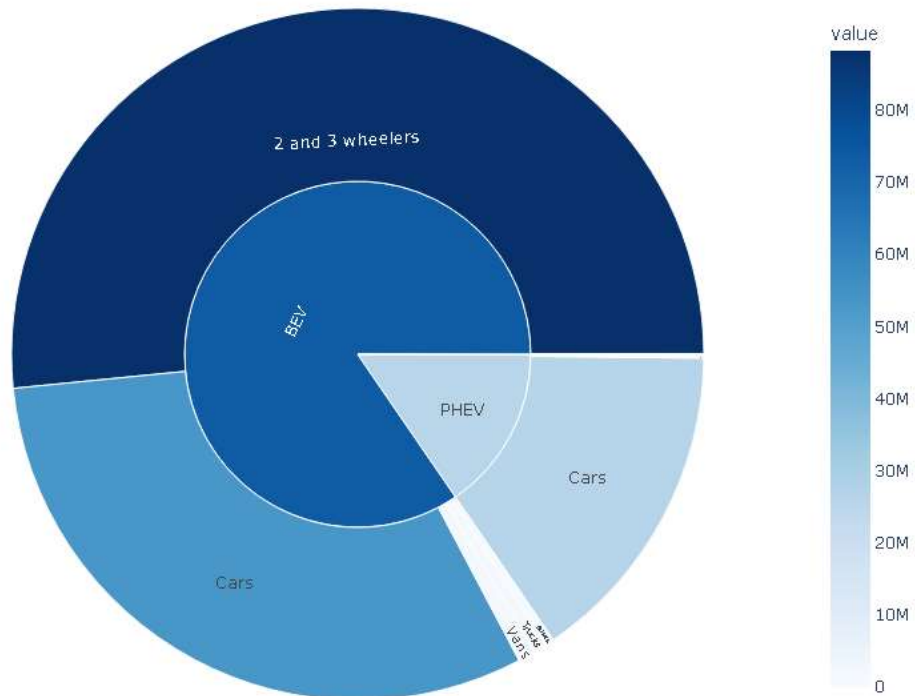
Global EV Market Share by Vehicle Segment



Battery Electric Vehicles Are Winning the Technology Race

How has the market transitioned between BEV, PHEV, and FCEV powertrains?

Powertrain Distribution Across Vehicle Segments



Key Insights

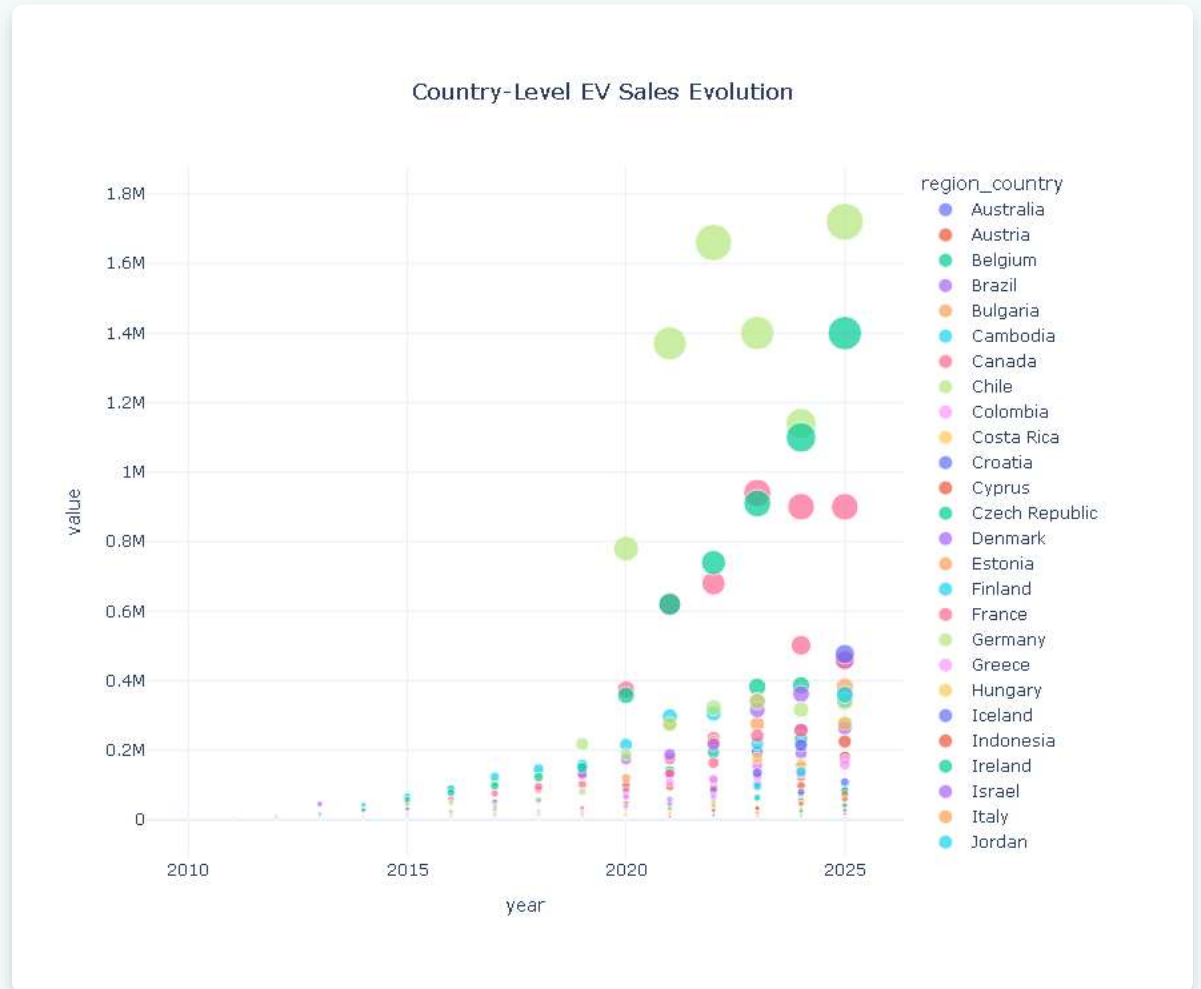
- ◆ Battery Electric Vehicles (BEVs) have become the dominant propulsion technology across most segments.
- ◆ Plug-in Hybrids (PHEVs) retain an important transitional role in several regional markets.
- ◆ Fuel Cell EVs (FCEVs) remain a niche technology with limited deployment to date.
- ◆ The trend reflects a progressive shift toward fully electric mobility rather than hybrid compromise solutions.

EV Adoption Is Geographically Concentrated

Which countries have emerged as the leading electric vehicle markets?

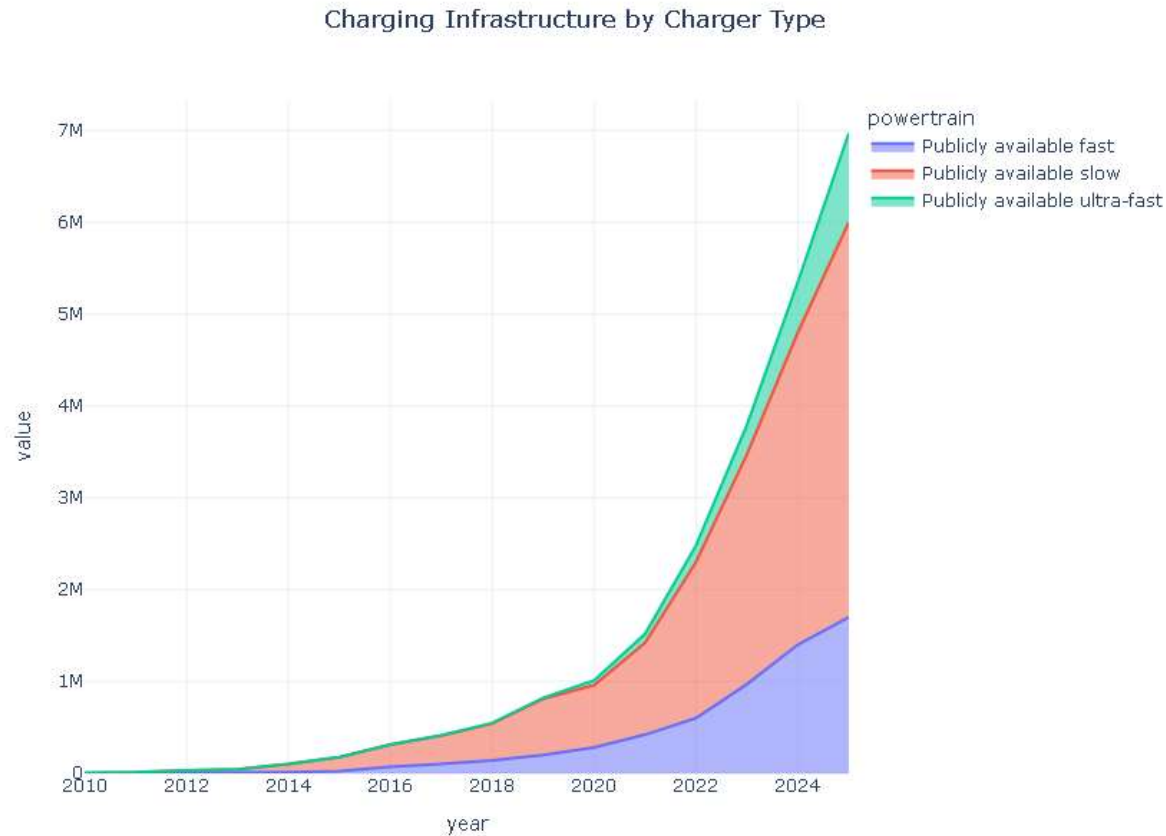
Key Insights

- ◆ A relatively small number of countries account for the majority of global EV sales.
- ◆ China, Europe, and North America dominate overall market activity.
- ◆ Several emerging markets show accelerating adoption despite lower cumulative totals.
- ◆ Geographic concentration persists, though diversification is increasing over time.



Charging Infrastructure Is Scaling With Vehicle Adoption

Has public charging infrastructure expanded at a pace comparable to EV adoption?



Key Insights

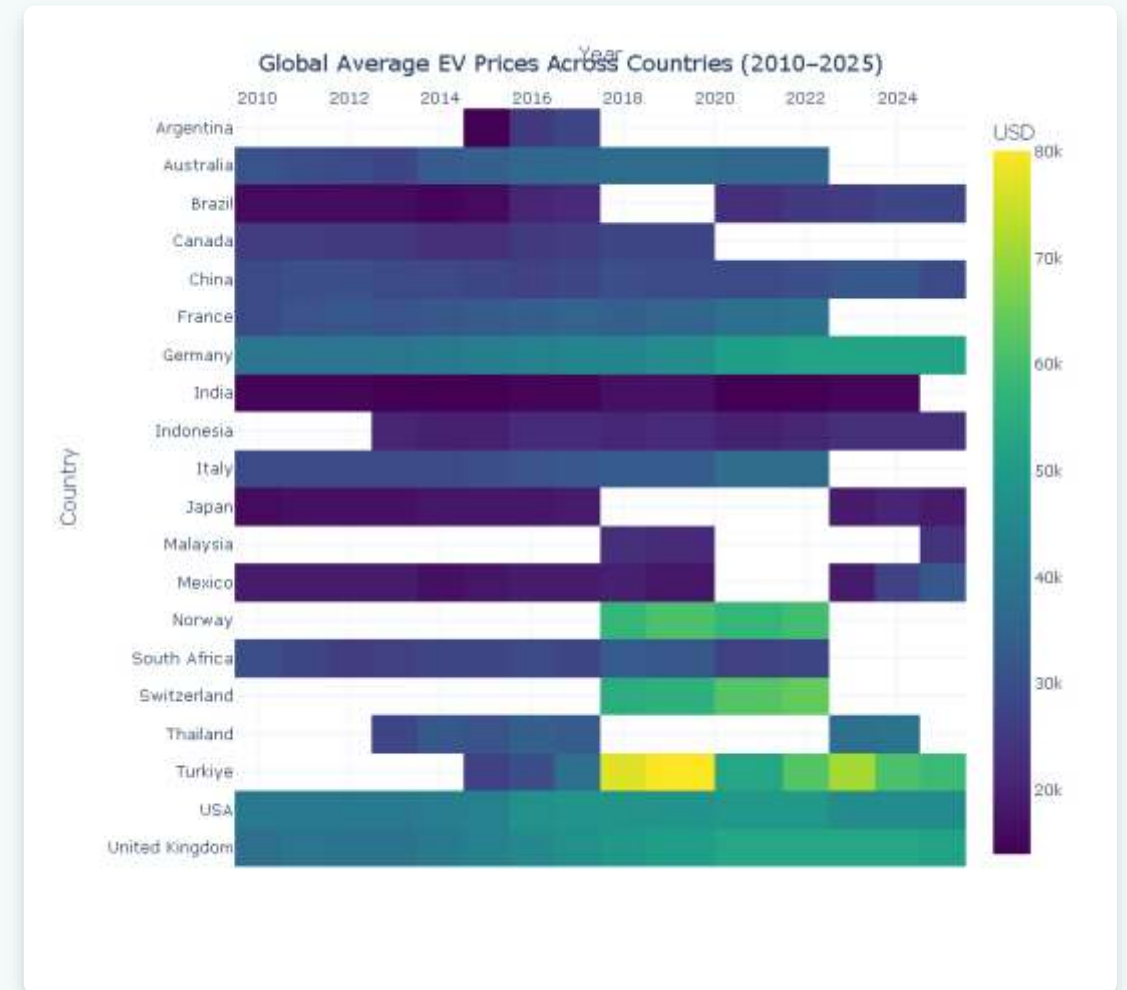
- ◆ Public charging stations increased consistently throughout the historical period.
- ◆ Infrastructure expansion accelerated after 2020, mirroring public and private investment surges.
- ◆ Fast-charging deployment is becoming increasingly important for long-distance electric mobility.
- ◆ Infrastructure growth tracks closely with passenger EV sales — an indicator of coordinated ecosystem development.

Affordability Varies Sharply Across Global Markets

How do average EV prices differ across countries and regions?

Key Insights

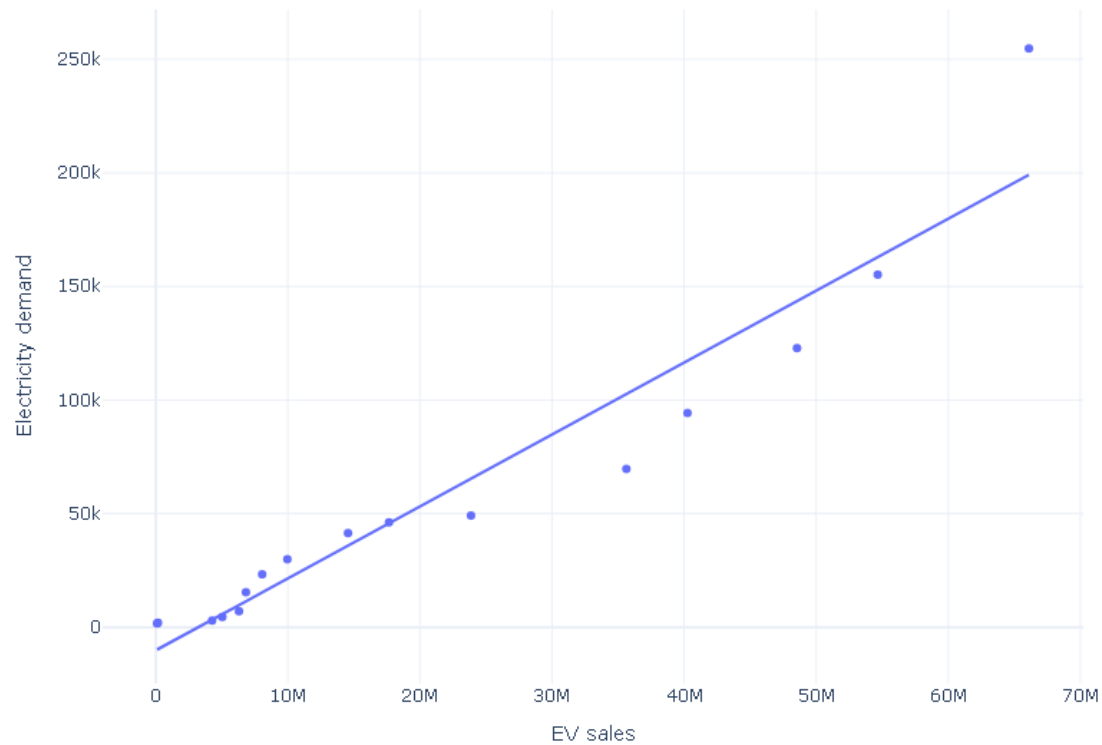
- ◆ Developed markets generally sustain higher average EV prices — larger batteries, premium models, stronger purchasing power.
- ◆ Several emerging markets show notably lower average prices, reflecting more affordable model availability.
- ◆ Regional price gaps confirm that affordability is shaped by taxation, local manufacturing, and competitive positioning — not global uniformity.
- ◆ Pricing strategy should stay localized rather than globally standardized.



The EV Ecosystem Moves Together

Which key EV ecosystem indicators exhibit the strongest relationships?

Relationship Between EV Sales and Electricity Demand



Key Insights

- ◆ EV sales and EV stock show a very strong positive relationship.
- ◆ Charging infrastructure expands in parallel with EV adoption — coordinated ecosystem growth.
- ◆ Electricity demand rises alongside fleet growth; battery deployment closely tracks EV stock.
- ◆ Oil displacement increases as EV adoption accelerates, reflecting the environmental payoff of electrification.

Executive KPI Dashboard — Historical Market (2010–2025)

A consolidated snapshot of global EV market growth, pricing, and infrastructure indicators for executive decision-making.



NOTEBOOK 03 · FORECASTING & STRATEGIC INSIGHTS

Part Two: The Road to 2035

Translating historical momentum into policy-scenario forecasts, infrastructure requirements, and investment priorities.

Two Policy Paths, One Direction: Massive Growth

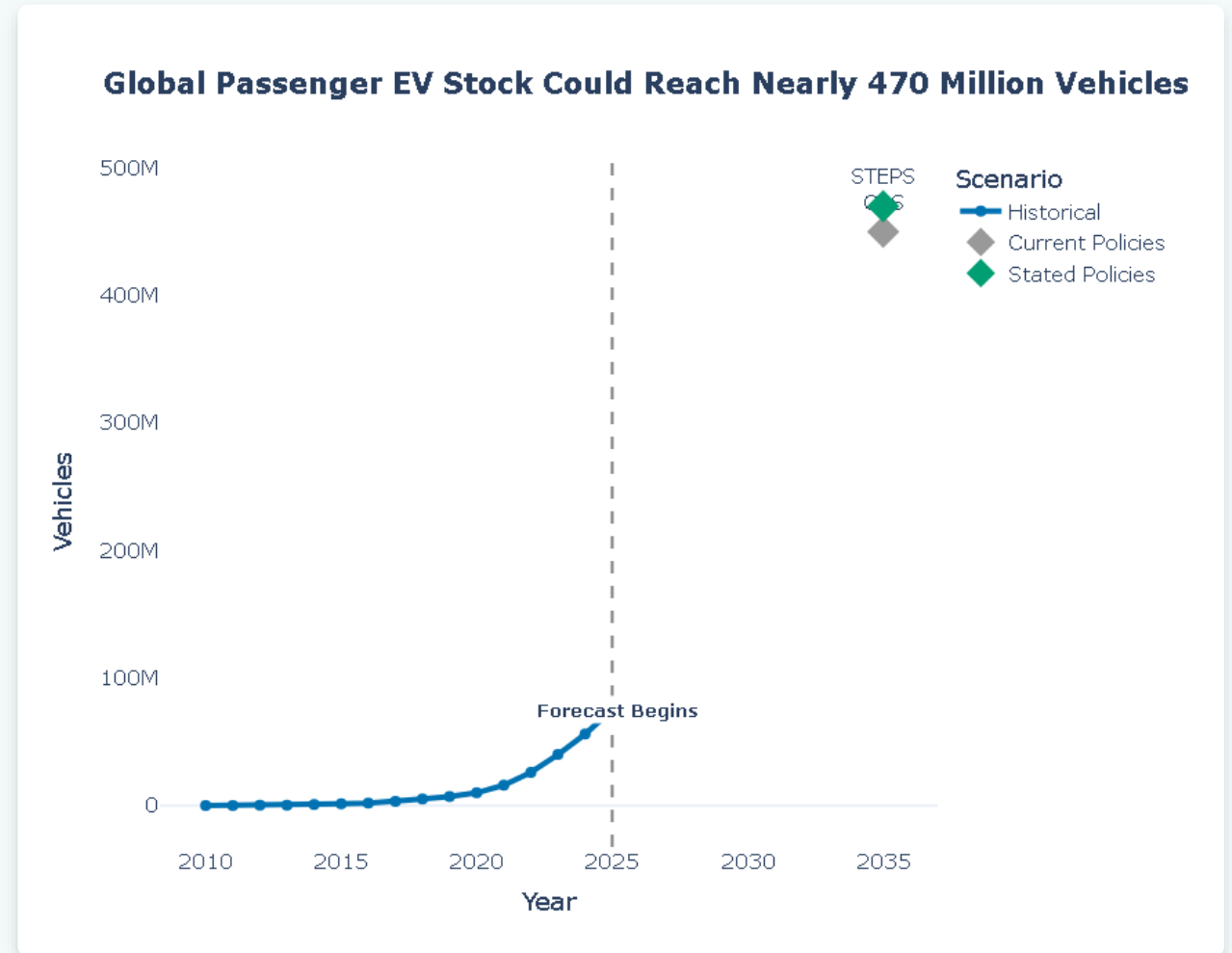
How large could the global passenger EV fleet become by 2035 under different policy scenarios?

75M → 450–470M

Passenger EV fleet, 2025 actual → 2035 projected

Key Insights

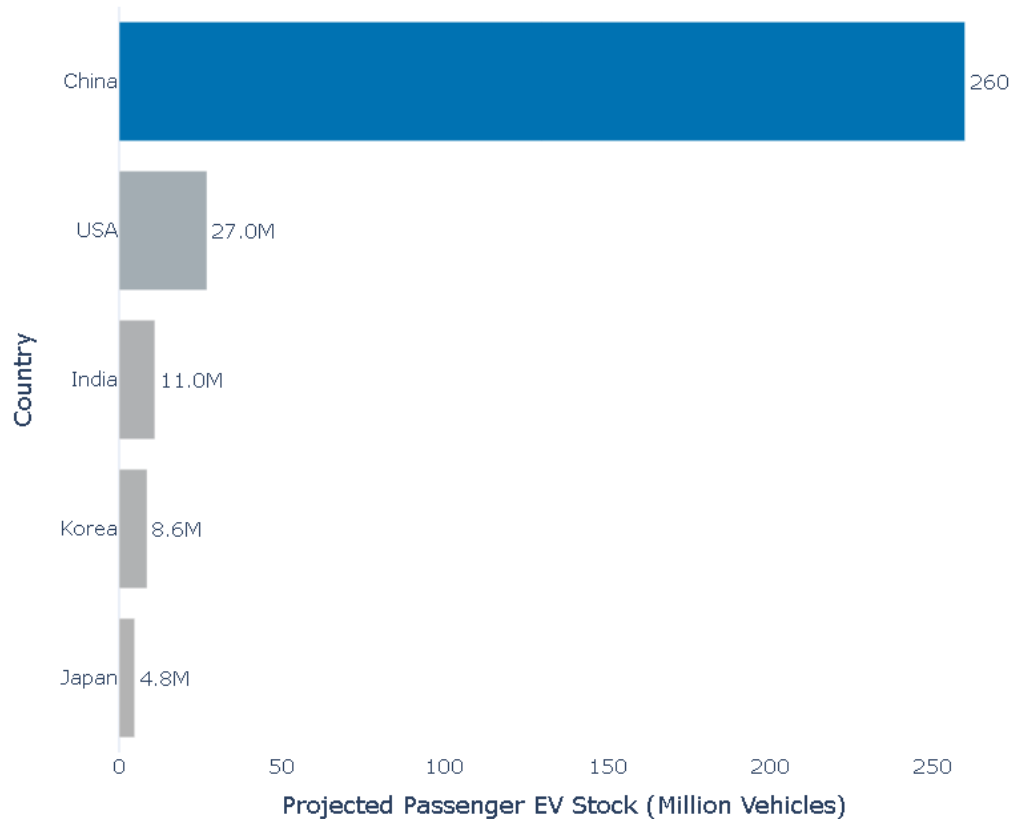
- ◆ Current Policies Scenario (CPS): fleet expands to ~450 million vehicles by 2035.
- ◆ Stated Policies Scenario (STEPS): fleet reaches ~470 million vehicles — 20 million more from stronger policy commitments.
- ◆ Both pathways confirm electric mobility will remain one of the fastest-growing transportation segments through the next decade.



Future Demand Will Concentrate in a Handful of Markets

Which countries are projected to lead the global passenger EV market by 2035 (STEPS)?

China Is Expected to Remain the World's Largest Passenger EV Market



Key Insights

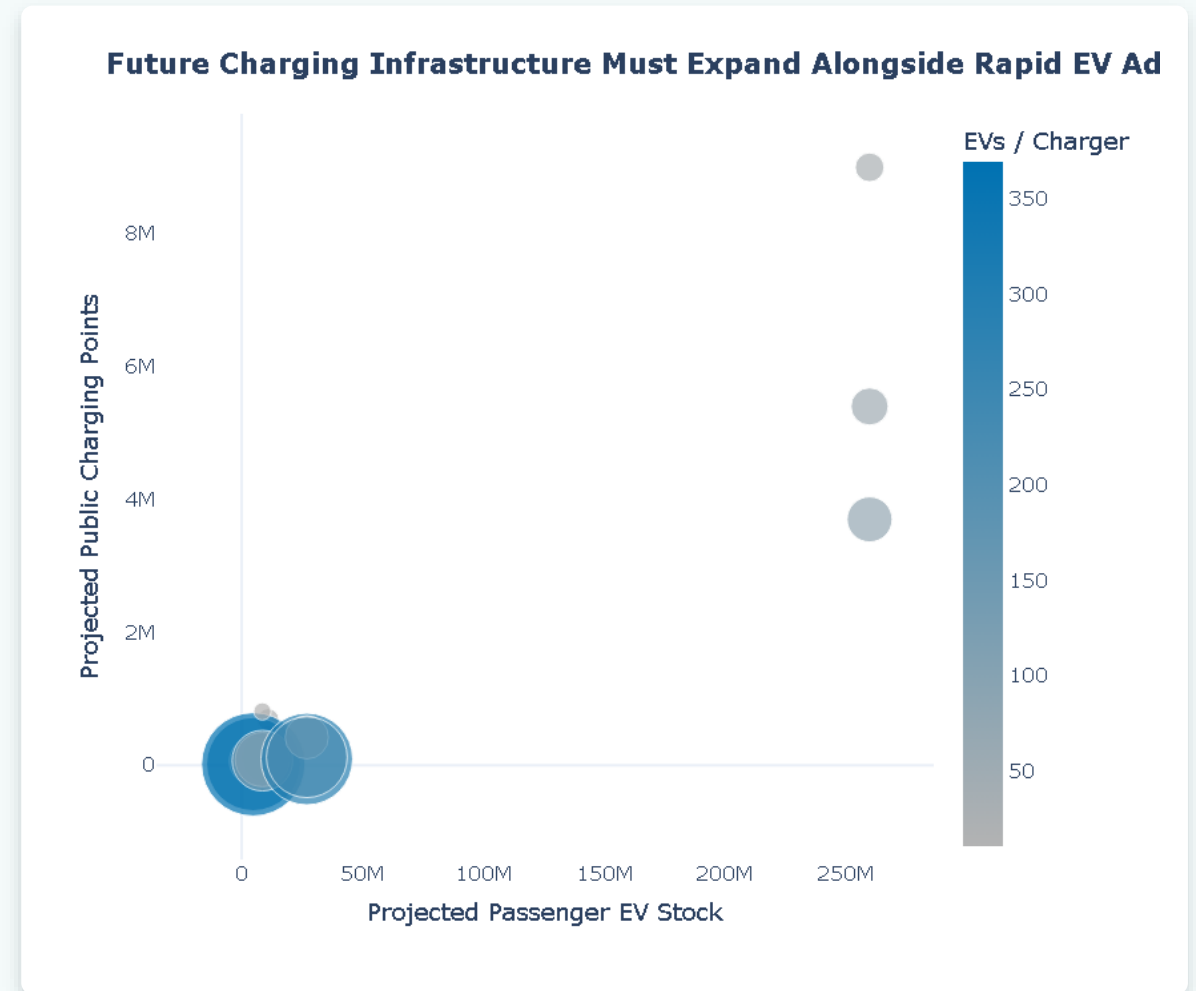
- ◆ A relatively small set of countries is projected to account for a large share of global EV deployment by 2035.
- ◆ Differences reflect market maturity, policy support, manufacturing capacity, and consumer adoption speed.
- ◆ Leading markets will require matching expansion in charging infrastructure, electricity generation, and battery supply chains.
- ◆ These markets represent the highest-priority locations for future ecosystem investment.

Infrastructure Must Keep Pace With the 2035 Fleet

Where will charging infrastructure need to expand the most by 2035?

Key Insights

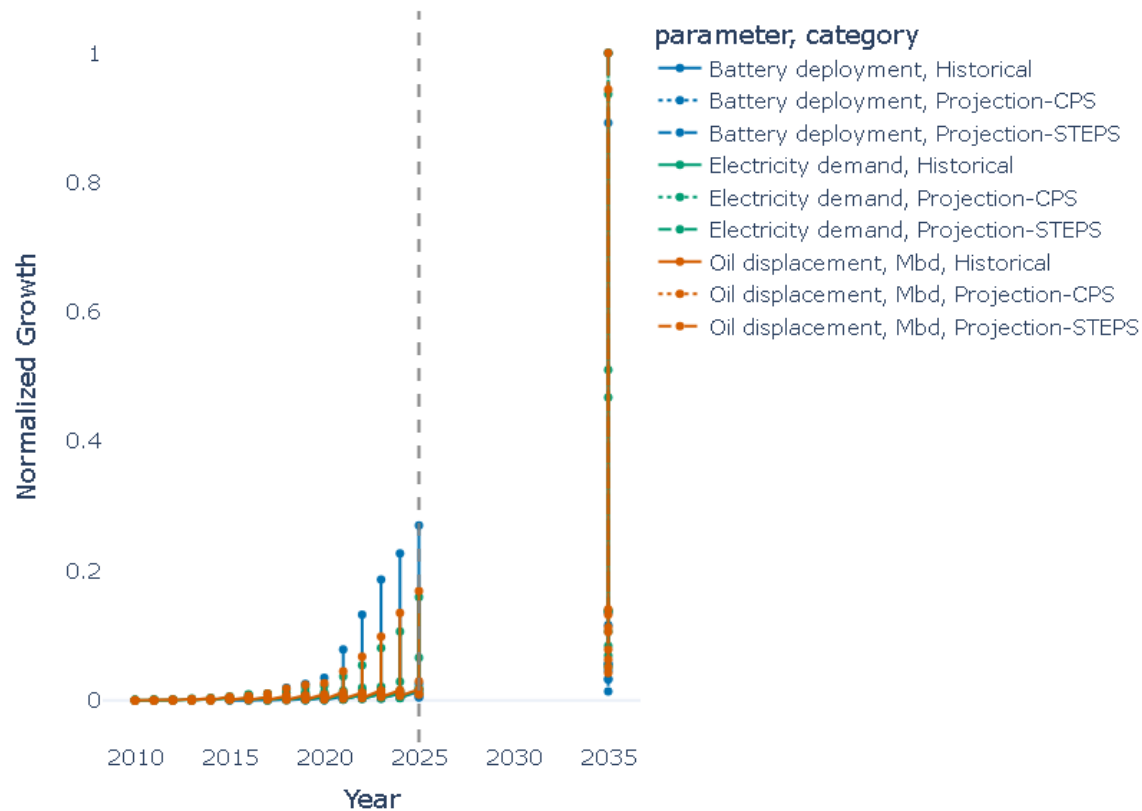
- ◆ Markets with large projected EV fleets require proportional expansion of public charging infrastructure.
- ◆ Countries with a high ratio of vehicles per charging point may face growing infrastructure pressure.
- ◆ These gaps represent clear investment opportunities for charging network operators and public-sector programs.
- ◆ Vehicle adoption and infrastructure deployment must progress together to sustain long-term growth.



Electrification Reshapes the Broader Energy System

How does projected EV growth align with battery deployment and oil displacement?

Future EV Adoption Is Expected to Reshape Global Energy Systems



Key Insights

- ◆ Future EV growth is projected to increase electricity demand and battery deployment in parallel.
- ◆ Oil displacement rises as adoption accelerates, reflecting reduced petroleum dependence.
- ◆ The transition extends well beyond transportation — into energy generation and industrial supply chains.
- ◆ Coordinated planning across transport and energy sectors will be essential to sustaining the trajectory.

Priorities for Governments, Industry & Capital

Governments	Manufacturers	Energy Utilities	Investors
● Strengthen long-term EV policy frameworks ● Accelerate public charging deployment ● Modernize electricity grid capacity	● Expand affordable passenger EV production ● Improve battery efficiency & cost ● Diversify offerings across segments	● Prepare grids for sustained EV demand growth ● Increase renewable integration ● Invest in smart-charging technology	● Prioritize high-growth, high-infrastructure-need markets ● Focus on charging networks & battery tech ● Track policy-driven demand signals

● FINAL CONCLUSION

A Structural, Long-Term Transformation

This analysis examined more than 49,000 observations covering global EV adoption between 2010 and 2025, and extended those findings through official IEA projections to 2035. The evidence points to sustained exponential growth in EV sales, expanding charging infrastructure, rising vehicle stock, and substantial policy-driven market transformation. Electrification is progressively broadening beyond passenger cars, and while affordability remains regionally heterogeneous, it continues to improve with technology and manufacturing scale.

450–470M

Projected 2035 passenger EV fleet

69.9%

Historical sales CAGR (2010–2025)

72

Countries in the analysis